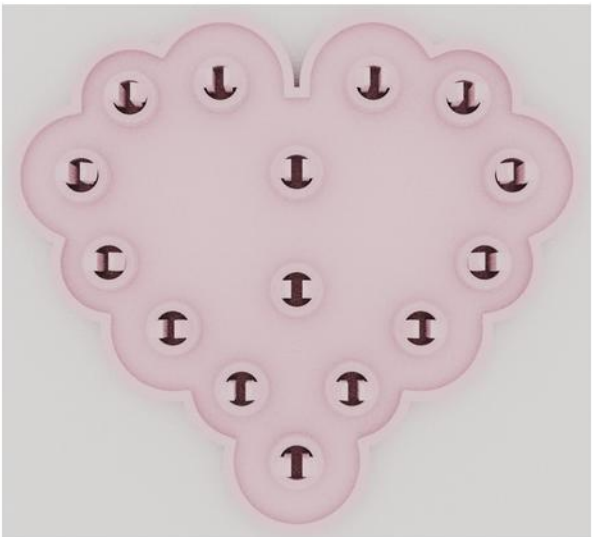
Gear Heart: printing

Two prints: the heart plate with its posts, and fifteen gears.

1 Print the plate



- **gear_heart_plate.3mf, flat, supports off. About 2 h 15 m and 40.5 g.**
- Fifteen posts stand up from it. Each has two springy prongs and a hook on top.

2 Print the gears



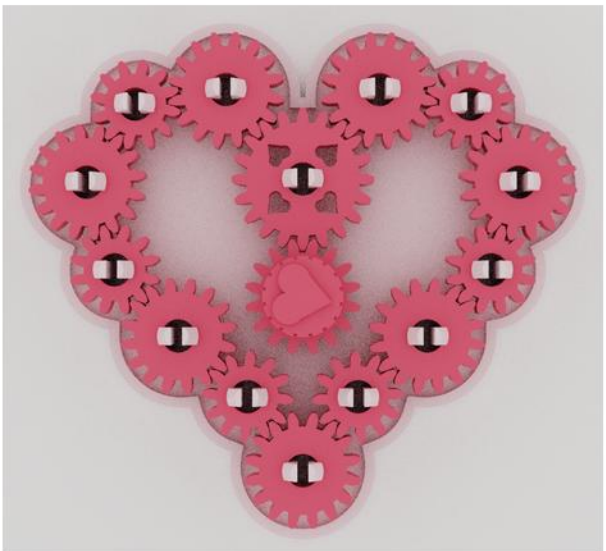
- **gear_heart_gears.3mf, flat, supports off. About 1 h 49 m and 26.6 g.**
- One big gear with heart holes, one knob gear, seven 16-tooth and six 12-tooth gears.

Design-review renders of the real parts, not photos. The colours are only for the pictures.

Gear Heart: putting it together

Design-review renders of the real parts, not photos.

3 Press the gears on, in this order



- **The heart-hole gear goes on the post in the top dip. The knob gear goes in the middle.**
- Then work down both sides from the top: a 16-tooth gear, then a 12, then a 16, and so on.
- Turn each gear until its teeth slip into its neighbour's, then press straight down until it clicks.
- The 16-tooth gear at the bottom point goes on last. Its teeth line up with both sides.

4 Turn the knob

- **Every gear turns together, each the opposite way to its neighbours.**

Good

- All fifteen gears turn smoothly from the knob.
- No gear lifts off as it turns.

Good to know

- The posts are made to be pressed on once.
- Pulling gears off and on again and again wears the hooks.

Design-review renders of the real parts, not photos. The colours are only for the pictures.